

HW Class12 Pt.2 (Population analysis) [Q13 Q14 BoxPlot]

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Population Scale Analysis

Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```
expr<-read.table  
"rs8067378_ENSG00000172057.6.txt"
```

```
[1] "rs8067378_ENSG00000172057.6.txt"
```

```
header = TRUE
```

```
url <- "https://bioboot.github.io/bggn213_W19/class-material/rs8067378_ENSG00000172057.6.txt"  
expr <- read.table(url, header = TRUE)
```

```
head(expr)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

```
# Sample size per genotype  
table(expr$geno)
```

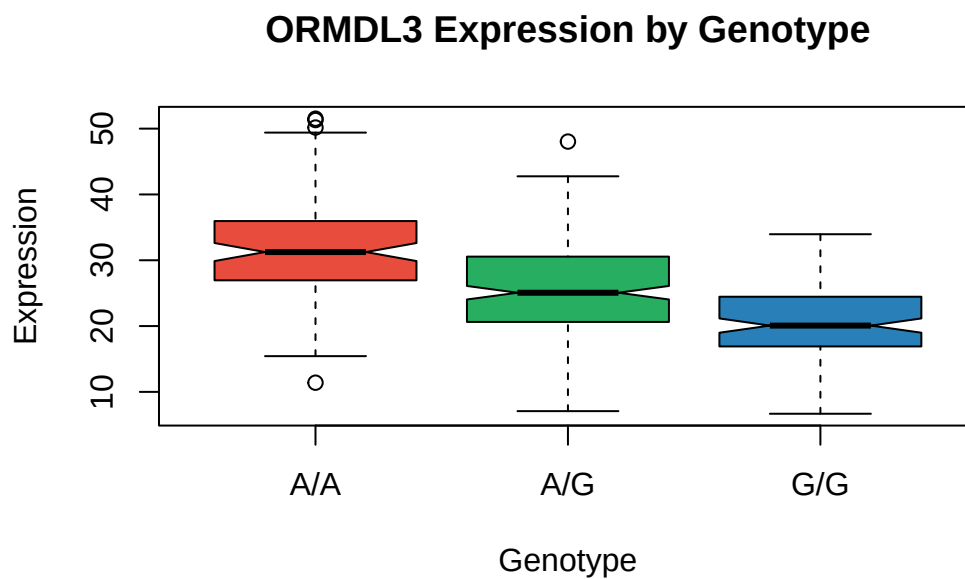
A/A A/G G/G
108 233 121

```
# Median expression per genotype  
tapply(expr$exp, expr$geno, median)
```

A/A A/G G/G
31.24847 25.06486 20.07363

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
boxplot(exp ~ geno, data = expr,  
        xlab = "Genotype",  
        ylab = "Expression",  
        col = c("#E74C3C", "#27AE60", "#2980B9"),  
        notch = TRUE,  
        main = "ORMDL3 Expression by Genotype")
```



Yes, the SNP appears to affect ORMDL3 expression. Individuals with the A/A genotype show higher expression levels than those with the G/G genotype.